COMPUTING AND INFORMATION SYSTEMS

Level 4 Diploma

UNIT 5 – WEB TECHNOLOGIES

JUNE 2014

Question 1

(a) Explain the differences between a relative and an absolute URL. (8 marks)

(b) Describe how a port number can be incorporated into a URL and discuss why this is important for some web applications. (12 marks)

Question 2

(a) Write HTML and CSS that will make text in a web document red, bold and italic. (8 marks)

(b) Explain how CSS can be incorporated into tags in order to control the layout and style of a web document. (12 marks)

Question 3

(a) Compare FOUR features of JSON and XML. (8 marks)

(b) Illustrate the process of delivering a web page from a web server to a client's web browser, incorporating the role of server-side scripting and providing annotation where necessary. (12 marks)

Question 4

Explain the process of setting up, installing and managing a web server. (20 marks)

Question 5

(a) Explain how to embed multimedia resources into HTML5. (8 marks)

(b) Discuss the concept of responsive web design, with reference to the varying capabilities of different user agents. (12 marks)

Question 6

(a) Review the key characteristics of a website that should be considered when selecting an appropriate web host. (8 marks)

(b) Contrast the design of a single-page application with the use of multiple server-generated pages. (12 marks)

Question 7

Discuss the advantages and disadvantages of client-side execution of code in a web browser. (20 marks)